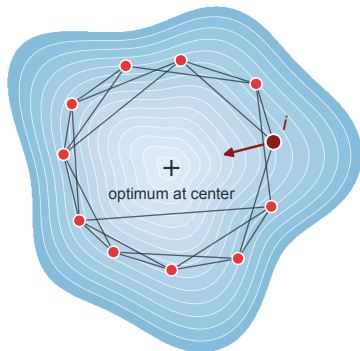
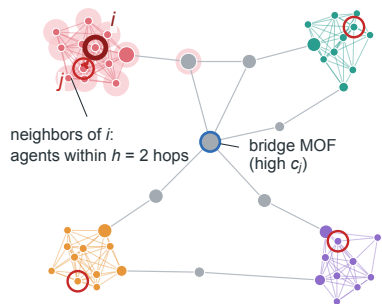


a SOCIAL (previous work)
continuous search space,
synthetic small-world agent graph

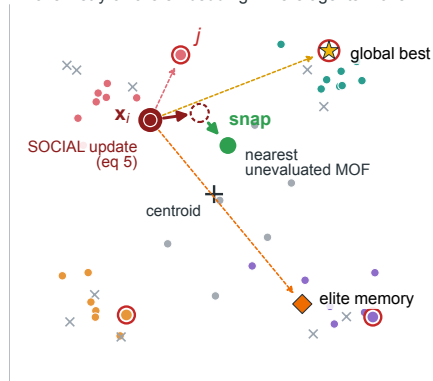


b Graph-SOCIAL (this study): the same agents in two views
MOF similarity network: who listens to whom

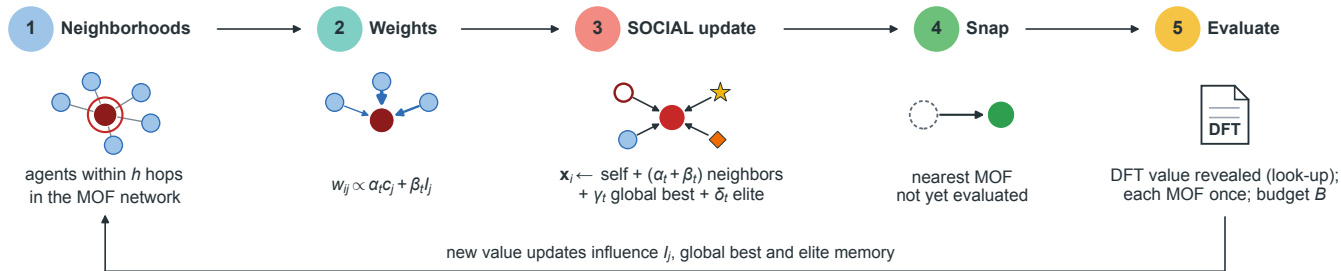


● MOF families ● connector MOF ○ agent × evaluated MOF ● 2-hop neighborhood of i

chemistry-aware embedding: where agents move



c One Graph-SOCIAL iteration (SOCIAL learning rules unchanged)



α_t, β_t decrease and γ_t, δ_t increase over iterations; worse-than-median agents may jump to the least-explored chemical community